

ArrayLists: Java

Java ArrayList

In Java, an `ArrayList` is used to represent a dynamic list. While Java arrays are fixed in size (the size cannot be modified), an `ArrayList` allows flexibility by being able to both add and remove elements.

```
// import the ArrayList package
import java.util.ArrayList;

// create an ArrayList called students
ArrayList<String> students = new
ArrayList<String>();
```

Modifying ArrayLists in Java

An `ArrayList` can easily be modified using built in methods.

To add elements to an `ArrayList`, you use the `add()` method. The element that you want to add goes inside of the `()`.

To remove elements from an `ArrayList`, you use the `remove()` method. Inside the `()` you can specify the index of the element that you want to remove.

Alternatively, you can specify directly the element that you want to remove.

```
import java.util.ArrayList;

public class Students {
    public static void main(String[] args) {

        // create an ArrayList called
        studentList, which initially holds []
        ArrayList<String>
        studentList = new ArrayList<String>();

        // add students to the ArrayList
        studentList.add("John");
        studentList.add("Lily");
        studentList.add("Samantha");
        studentList.add("Tony");

        // remove John from the ArrayList,
        then Lily
        studentList.remove(0);
        studentList.remove("Lily");

        // studentList now holds [Samantha,
        Tony]

        System.out.println(studentList);
    }
}
```

